Linear Algebra Exam Answers (Concise Exam-Room Version)

1. Compute the determinant

$$D_n = \begin{vmatrix} a_1+\lambda & a_2 & \cdots & a_n \\ a_1 & a_2+\lambda & \cdots & a_n \\ \vdots & \vdots & & \vdots \\ a_1 & a_2 & \cdots & a_n+\lambda \end{vmatrix}$$

Solution:

Add columns $2, 3, \ldots, n$ all to column 1 and factor out the common factor $\lambda + \sum_{i=1}^{n} a_i$:

$$D_n = \left(\lambda + \sum_{i=1}^{n} a_i\right) \begin{vmatrix} 1 & a_2 & \cdots & a_n \\ 1 & a_2+\lambda & \cdots & a_n \\ \vdots & \vdots & & \vdots \\ 1 & a_2 & \cdots & a_n+\lambda \end{vmatrix}$$

Subtract row 1 from row i ($i = 2, \ldots, n$), reducing to upper triangular form:

$$D_n = \left(\lambda + \sum_{i=1}^{n} a_i\right) \begin{vmatrix} 1 & a_2 & \cdots & a_n \\ 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \lambda \end{vmatrix}$$

$$\boldsymbol{D_n = \lambda^{n-1}(\lambda + a_1 + a_2 + \cdots + a_n)}$$

2. Vector operations

Given $\boldsymbol{\alpha} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$

Solution:

$$1.\ \alpha^T\alpha = \begin{pmatrix} 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = 1 + 4 + 1 = \mathbf{6}$$

$$2.\ \alpha\alpha^T = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

$$3.\ \text{By the associative law: } \alpha\alpha^T\alpha = \alpha(\alpha^T\alpha) = 6\alpha = \begin{pmatrix} 6 \\ 12 \\ 6 \end{pmatrix}$$

3. Prove that $A - E$ is invertible and find its inverse

Given $A^2 - 5A + E = 0$

Proof:

From $A^2 - 5A + E = 0$, rearranging gives:

$$A^2 - A - 4A + 4E = 3E$$

$$(A - E)(A - 4E) = 3E$$

i.e.

$$(A - E) \cdot \frac{1}{3}(A - 4E) = E$$

By the definition of invertibility, $A - E$ is invertible, and

$$\boldsymbol{(A - E)^{-1} = \frac{1}{3}(A - 4E)}$$

4. Discussion of the solutions of the system of linear equations

$$\begin{cases} ax_1 + x_2 + x_3 = 0 \\ x_1 + ax_2 + x_3 = 2 \\ x_1 + x_2 + ax_3 = a \end{cases}$$

Solution:

The determinant of the coefficient matrix

$$|A| = \begin{vmatrix} a & 1 & 1 \\ 1 & a & 1 \\ 1 & 1 & a \end{vmatrix} = (a + 2)(a - 1)^2$$

(1)

(2) **No solution**: when $a = 1$, the augmented matrix

$$\overline{A} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$r(A) = 1 \neq r(\overline{A}) = 2$, so there is no solution when $\boldsymbol{a = 1}$.

(3) **Infinitely many solutions**: when $a = -2$,

$$\overline{A} = \begin{pmatrix} -2 & 1 & 1 & 0 \\ 1 & -2 & 1 & 2 \\ 1 & 1 & -2 & -2 \end{pmatrix} \to \begin{pmatrix} 1 & -2 & 1 & 2 \\ 0 & -3 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$r(A) = r(\overline{A}) = 2 < 3$, so there are infinitely many solutions.

The equivalent system:

$$\begin{cases} x_1 - 2x_2 + x_3 = 2 \\ -3x_2 + 3x_3 = 4 \end{cases}$$

Let the free variable be $x_3 = c$; solving gives $x_2 = c - \frac{4}{3}$ and $x_1 = c - \frac{2}{3}$.

The general solution:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \boldsymbol{c} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} -\frac{2}{3} \\ \frac{4}{3} \\ 0 \end{pmatrix} \quad (\boldsymbol{c} \in \mathbb{R})$$

5. Prove that $\beta_1, \beta_2, \beta_3$ are linearly independent

Given that $\alpha_1, \alpha_2, \alpha_3$ are linearly independent, and $\beta_1 = \alpha_1 + \alpha_2$, $\beta_2 = \alpha_2 + \alpha_3$, $\beta_3 = \alpha_3 + \alpha_1$

Proof:

Set $k_1\beta_1 + k_2\beta_2 + k_3\beta_3 = 0$; substituting gives:

$$(k_1 + k_3)\alpha_1 + (k_1 + k_2)\alpha_2 + (k_2 + k_3)\alpha_3 = 0$$

Since $\alpha_1, \alpha_2, \alpha_3$ are linearly independent,

$$\begin{cases} k_1 + k_3 = 0 \\ k_1 + k_2 = 0 \\ k_2 + k_3 = 0 \end{cases}$$

The determinant of the coefficients

$$\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 2 \neq 0$$

The system has only the zero solution $k_1 = k_2 = k_3 = 0$, so $\beta_1, \beta_2, \beta_3$ are linearly independent.

6. Determine the parameters and find the maximal linearly independent subset

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 3 & 2 & 1 & -3 & a \\ 0 & 1 & 2 & 6 & 3 \\ 5 & 4 & 3 & -1 & b \end{pmatrix}$$

Solution:

Apply elementary row operations to A:

$$A \to \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & -1 & -2 & -6 & a-3 \\ 0 & 0 & 0 & 0 & a \\ 0 & 0 & 0 & 0 & b-a-2 \end{pmatrix}$$

From $r(A) = 2$ we get $\begin{cases} a = 0 \\ b - a - 2 = 0 \end{cases}$, i.e. $\boldsymbol{a = 0,\ b = 2}$.

The reduced row echelon form is then:

$$A \to \begin{pmatrix} 1 & 0 & -1 & -5 & -2 \\ 0 & 1 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

- A maximal linearly independent subset of the set of column vectors: $\boldsymbol{\alpha_1, \alpha_2}$

- Linear expressions of the remaining vectors:

$$\boldsymbol{\alpha_3} = -\boldsymbol{\alpha_1} + 2\boldsymbol{\alpha_2}, \quad \boldsymbol{\alpha_4} = -5\boldsymbol{\alpha_1} + 6\boldsymbol{\alpha_2}, \quad \boldsymbol{\alpha_5} = -2\boldsymbol{\alpha_1} + 3\boldsymbol{\alpha_2}$$

7. Find the eigenvalues and eigenvectors

(1) $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Solution:

The characteristic equation:

$$|\lambda E - A| = \begin{vmatrix} \lambda - 1 & -1 & 0 \\ -1 & \lambda - 1 & 0 \\ 0 & 0 & \lambda - 1 \end{vmatrix} = \lambda(\lambda - 1)(\lambda - 2) = 0$$

The eigenvalues are $\boldsymbol{\lambda_1 = 0,\ \lambda_2 = 1,\ \lambda_3 = 2}$.

- $\lambda_1 = 0$: solve $-A\boldsymbol{x} = 0$; the fundamental system of solutions is $\boldsymbol{\xi_1} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$, and the eigenvectors are $k_1\xi_1$ $(k_1 \neq 0)$.
- $\lambda_2 = 1$: solve $(E - A)\boldsymbol{x} = 0$; the fundamental system of solutions is $\boldsymbol{\xi_2} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$, and the eigenvectors are $k_2\xi_2$ $(k_2 \neq 0)$.
- $\lambda_3 = 2$: solve $(2E - A)\boldsymbol{x} = 0$; the fundamental system of solutions is $\boldsymbol{\xi_3} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, and the eigenvectors are $k_3\xi_3$ $(k_3 \neq 0)$.

(2) $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

Solution:

The characteristic equation:

$$|\lambda E - A| = \begin{vmatrix} \lambda - 1 & -1 \\ -1 & \lambda - 1 \end{vmatrix} = \lambda(\lambda - 2) = 0$$

The eigenvalues are $\boldsymbol{\lambda_1 = 0,\ \lambda_2 = 2}$.

- $\lambda_1 = 0$: solve $-A\boldsymbol{x} = 0$; the fundamental system of solutions is $\boldsymbol{\xi_1} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$, and the eigenvectors are $k_1\xi_1$ $(k_1 \neq 0)$.
- $\lambda_2 = 2$: solve $(2E - A)\boldsymbol{x} = 0$; the fundamental system of solutions is $\boldsymbol{\xi_2} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, and the eigenvectors are $k_2\xi_2$ $(k_2 \neq 0)$.